A characteristic outflow posed on the relaxed state

Notes for a discussion of NSCBC in flashing, relaxing two-phase flow

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Abstract

We are simulating CO_2 pipeline depressurisation with a six-equation two-phase model (Pelanti–Shyue) on real-fluid equations of state (Peng–Robinson and GERG-2008), whose closures are an instantaneous pressure-equilibrium constraint between the phases, finite-rate thermal and mass-transfer relaxation, and nucleation (flash). The interior scheme is wave propagation. At a subsonic outflow the classical NSCBC construction, written on the frozen (Wallis) eigenstructure, reflects a large fraction of any wave that has been relaxed: the relaxed wave has components on both frozen characteristics, and any condition that sets the entering frozen wave from the far field returns part of the leaving relaxed one. We built an alternative that treats the strength of the entering discrete wave as the one modelled quantity and poses the boundary condition on the state the cell relaxes to at the end of the step, with the relaxed impedance measured from the trial evaluations themselves. On plane relaxed waves it takes the reflection from 0.24 to 0.01 with nothing lost on pure vapour. On the production outlet — a supersonic two-phase jet core impinging on the outflow plane with the surrounding face in reversed flow — it has taught us, run by run, which parts of the classical construction were doing what. This note frames the construction against Poinsot–Lele for a reader whose reference application is reacting flow, and lists the questions we would like to discuss.

1 Where the applications overlap

The reacting-flow and the flashing-flow outlet share the structure that made NSCBC necessary in the first place. Both have a subsonic outflow through which acoustic waves must leave without reflection while the mean pressure is held to a far-field value; both carry a convected scalar field (composition; volume fraction and phase state) that must cross the boundary without generating pressure; both have a stiff source term whose time scale is far below the acoustic one. Sutherland and Kennedy's observation for reacting flows — that the source belongs *in* the boundary relation, because the characteristic amplitudes are not decoupled from a stiff source over a time step — is the starting point here too.

Both applications also meet the two situations the literature treats least well. Reversed flow on part of the outlet (a recirculating jet, a Mach-disk recompression) turns an outflow face into a partial inflow with the entering entropy and vorticity waves undetermined. And the finite-volume setting has no boundary node: the condition must be imposed through a ghost state or on a face flux, so the drift and the near-sonic ill-conditioning that Masset et al. analyse for ghost-cell CBC are shared.

2 The model, in the variables used below

The six-equation model of Pelanti and Shyue carries, for two phases $k = 1$ (liquid) and $k = 2$ (vapour, or supercritical fluid) sharing one velocity $\mathbf{u}$, the volume fraction α_1 ($\alpha_2 = 1 - \alpha_1$), the partial masses $m_k = \alpha_k \rho_k$, the mixture momentum $\rho\mathbf{u}$ and the phase total energies $\mathcal{E}_k = \alpha_k \rho_k E_k$ with $E_k = e_k + \frac{1}{2}|\mathbf{u}|^2$. Each phase has its own density ρ_k, internal energy e_k, temperature T_k

and pressure $P_k = P_k(\rho_k, e_k)$ from its own branch of the equation of state; the mixture pressure is $P = \alpha_1 P_1 + \alpha_2 P_2$ and the mixture density $\rho = m_1 + m_2$. The system is

$$\partial_t \alpha_1 + \mathbf{u}\cdot\nabla\alpha_1 = \mathcal{S}_\alpha, \tag{1}$$

$$\partial_t m_k + \nabla\cdot(m_k\mathbf{u}) = \pm\Gamma, \tag{2}$$

$$\partial_t(\rho\mathbf{u}) + \nabla\cdot(\rho\mathbf{u}\otimes\mathbf{u} + P\mathsf{I}) = 0, \tag{3}$$

$$\partial_t \mathcal{E}_k + \nabla\cdot(\mathcal{E}_k\mathbf{u} + \alpha_k P_k\mathbf{u}) = \mp P_I\,\partial_t\alpha_1 \pm \mathcal{Q} \pm \Gamma H_I, \tag{4}$$

with $\mathcal{Q}$ the thermal relaxation ($T_1 \to T_2$), Γ the mass transfer (driving the Gibbs energies together; nucleation of a phase that is absent is a separate flash operator), the interfacial pressure $P_I = \frac{1}{2}(P_1 + P_2)$, and $\mathcal{S}_\alpha$ the term that couples the two phase pressures. In Pelanti and Shyue's formulation $\mathcal{S}_\alpha = \mu\,(P_1 - P_2)$ with $\mu \to \infty$; we take that limit literally rather than as a rate. The hyperbolic step advances the six equations with α_1 advected and each phase carrying its own pressure, and after every step α_1 is reset, cell by cell, to the value at which $P_1 = P_2$ (a Newton on α_1, with the interfacial work $\mp P_I\,\delta\alpha_1$ deposited in the phase energies as the discrete form of the non-conservative term). Pressure equality is therefore a constraint of the closure — mechanical equilibrium, imposed by redistributing volume between the phases — and not a relaxation process with a time scale: a finite-rate version was measured inert and removed. Thermal and mass-transfer relaxation are finite-rate in the production closure (the coupled "X3" operator, which integrates both on the $P_1 = P_2$ manifold and whose fixed point is the equilibrium state). The mixture total energy $\rho E = \mathcal{E}_1 + \mathcal{E}_2$ is fluxed in strict conservation form; the volume-fraction and phase-energy equations are not in divergence form, which is why the interior uses wave propagation rather than flux differencing.

The phases are evaluated on real-fluid equations of state: Peng–Robinson (analytic, and a tabulated form) and GERG-2008 (the reference multiparameter equation, tabulated), each with a branch-locked liquid/vapour inversion so that a phase is always queried on its own branch and "no root" is reported rather than substituted. Every saturation quantity the closures use ($P_{\mathrm{sat}}(T)$, $T_{\mathrm{sat}}(P)$, the saturated phase states) is the equation of state's own coexistence curve.

The homogeneous-equilibrium limit. As all three relaxation times go to zero, $P_1 = P_2$, $T_1 = T_2$ and $g_1 = g_2$ at every instant; α_1, m_1 and m_2 cease to be independent and are fixed by (ρ, e) through the saturation curve, and the six equations collapse to the three-equation Euler system for the mixture with an equilibrium (HEM) equation of state $P = P_{\mathrm{HEM}}(\rho, e)$. The HEM limit is what the closures relax toward and what our exact two-phase reference solutions are computed on; the finite-rate model sits between the frozen system and that limit.

Frozen and relaxed sound speeds. Throughout this note "frozen" means that the relaxation terms are ignored: the sources $\mathcal{S}_\alpha$, $\mathcal{Q}$ and Γ are set to zero, so each phase keeps its own pressure, temperature and mass, and the mixture is two phases compressed side by side. Three sound speeds follow from which of the closures act on a disturbance. With none of them acting — the frozen system — each phase is compressed along its own isentrope at fixed volume fraction, the mixture's bulk modulus is the mass-weighted sum of the phase moduli, and the speed is $c_f^2 = Y_1 c_1^2 + Y_2 c_2^2$ with $Y_k = m_k/\rho$ (Wallis). In the model as we solve it no physical wave travels at c_f: the pressure constraint is re-imposed after every step, so c_f is the speed of the characteristics of the hyperbolic operator within a step, the object the split scheme propagates and the Riemann solver is built on, and nothing more. With the pressure constraint acting, the volume fractions redistribute so that the more compressible phase takes more of the volume change; the mixture's compressibility is the volume-weighted sum of the phase compressibilities, $1/(\rho c_w^2) = \sum_k \alpha_k/(\rho_k c_k^2)$, and the speed c_w (Wood) satisfies $c_w \le c_f$, with the gap large in a bubbly mixture (here $c_w = 174$ m/s against $c_f = 282$ m/s); this is the speed of a wave in the model with the thermal and mass-transfer legs switched off. With those legs also in equilibrium,

evaporation and condensation along the wave absorb part of the pressure change and the speed is lower still: c_{eq}, the sound speed of the homogeneous equilibrium mixture. Two further symbols are used below. The *impedance* $Z = \rho c$ is the ratio $\Delta P/\Delta u$ across an acoustic wave travelling at speed c, so $Z_f = \rho c_f$ for a frozen wave and $Z_{\mathrm{rel}} = \rho\, c_{\mathrm{rel}}$ for a relaxed one; and c_{rel} denotes the speed at which a wave actually travels in the relaxed flow — c_w when only the pressure constraint has acted, c_{eq} when the thermal and mass-transfer legs are in equilibrium too, and in between for the finite-rate closure (139 m/s measured for the production closure on the mixture above). The scheme propagates frozen characteristics within the step and applies the constraint and the relaxation legs afterwards; the physical waves travel at c_{rel}.

3 Where flashing flow differs from reacting flow

Three differences matter at the boundary, and they compound.

The source changes the eigenstructure. A reaction term moves the state along the composition direction and leaves the acoustic eigenstructure of the gas essentially alone. Relaxation in a two-phase mixture does not: the frozen (Wallis) sound speed of the six-equation system,

$$c_f^2 = Y_1 c_1^2 + Y_2 c_2^2, \tag{5}$$

is the speed of the characteristics the scheme propagates, while the speed at which a wave actually travels once the pressure constraint has acted is the Wood speed,

$$\frac{1}{\rho\, c_w^2} = \frac{\alpha_1}{\rho_1 c_1^2} + \frac{\alpha_2}{\rho_2 c_2^2}, \tag{6}$$

and after thermal and mass-transfer relaxation the equilibrium speed c_{eq}, lower still. The subcharacteristic condition $c_{\mathrm{eq}} \le c_w \le c_f$ is what makes the split scheme stable, and it is also what makes the boundary reflect. A relaxed wave travelling at c_{rel} carries $\Delta P/\Delta u = Z_{\mathrm{rel}} = \rho\, c_{\mathrm{rel}}$; the frozen Riemann invariants of the boundary cell, $R^{\pm} = u \pm P/Z_f$, are built with $Z_f = \rho c_f$, so that same wave has a non-zero component on the *entering* frozen characteristic,

$$\Delta R^{-} = \Delta P\Big(\frac{1}{Z_{\mathrm{rel}}} - \frac{1}{Z_f}\Big), \tag{7}$$

which any condition written on the frozen characteristics sends back into the domain. The reflected fraction of the pressure amplitude is the impedance mismatch $(Z_f - Z_{\mathrm{rel}})/(Z_f + Z_{\mathrm{rel}})$. For our saturated mixture at $\alpha_1 = 0.05$, $c_f = 282$ m/s and the measured $c_{\mathrm{rel}} = 139$ m/s give a predicted reflection of about 0.3; measured 0.22–0.24, the same for the NSCBC ghost and for a plain zero-gradient copy. Nothing in the invariant construction is at fault; the basis is.

The source acts along the wave, not only in the cell. A depressurising liquid expands through an isentropic fan along which it crosses the saturation dome and flashes; the fan's own sound speed collapses as vapour forms, and the physical exit chokes at the equilibrium sound speed, far below the frozen one. Our boundary construction for a venting face follows exactly that path. Starting from the phase states of the boundary cell, it marches the pressure from the cell value toward the ambient value in substeps (each limited to a 5 % density change of either phase — a march in pressure along the isentrope, not a march in time through the CFD step), moves each phase along its own isentrope at every substep, and, whenever the mixture enthalpy at the current pressure lies inside the saturation dome, re-closes the mixture to the equilibrium (HEM) saturated state at that pressure, which is where the flash enters and where the path's sound speed falls. The march stops at the ambient pressure or at the point of maximal mass flux ρu along the path, the critical-flow condition; the state reached is the ghost state. We measure

this construction on a one-dimensional tube of supercritical liquid CO_2 (100 bar, 320 K) venting through one end to 1 bar, scoring the mass that has left the tube by 80 μs. The reference is the same tube computed on a domain twice as long, with the vent plane an interior interface and 1 bar vapour beyond it, so that no boundary model is involved; the fan construction vents 4.4 % more mass than that reference. A relaxation operator that acts on the boundary cell *after* a frozen wave has entered cannot reproduce this. The wave that enters the cell is a frozen HLLC jump; a frozen jump from 100 bar to about 1 bar on a liquid is an isentropic expansion by 99 bar with no phase change, and the liquid branch of the equation of state has no state there, so the trial evaluation refuses it. The physical path flashes on the way down, with the pressure falling through the dome in a two-phase mixture, which is what the substepped march follows and a single jump cannot. This is the one place the new construction has had to concede ground, and it is one of the questions below.

The boundary state has a thermodynamic admissibility problem. Each phase is evaluated on its own branch of a cubic equation of state, and a boundary value obtained by extrapolation or by construction may fall outside the region where that phase-locked evaluation has a solution — a liquid density and energy for which the liquid branch has no root, say — and is then simply not a state. Masset et al. meet the same problem for multicomponent gases with complex state laws and answer it by shrinking the boundary gradient until the ghost becomes admissible. Our implementation avoids floors and thresholds of that kind: where a construction generates a non-physical value we do not adjust it, we discard it, count the refusal, and fall back to a lower-accuracy construction that is admissible by design. For example, an extrapolated ghost state whose phases have no root on their branches is replaced by a copy of the boundary cell, which is admissible because the boundary cell is; and a trial state of the relaxed boundary that refuses is replaced by the wave the interior fan had already sent.

4 The interior scheme, and the boundary face in its language

The interior is Berger–LeVeque wave propagation. At every face the HLLC fan gives three waves $\mathcal{W}_l$ with speeds s_l, and the two cells that share the face are each updated with the sum of the fluctuations $\sum s_l \mathcal{W}_l$ of the waves travelling into them, i.e. arriving from upwind; a limited second-order correction flux is added on top. The non-conserved quantities — the volume fraction and the two phase energies — are updated by those same fluctuations rather than by a flux divergence, which would require a carefully synchronised subtraction of the non-conservative source term. The viscous contribution is a Newtonian stress with a constant effective viscosity μ (a deck parameter chosen to give the orifice shear layer a resolved thickness), added in flux form with the viscous heating partitioned to the phase energies by mass fraction. After the hydrodynamic increment, a reaction operator $\mathcal{R}$ acts cell by cell: the mixture-consistency resyncs, the phase-removal operators (a phase whose volume fraction or partial mass has fallen below what the model can resolve is removed and its mass and energy assigned to the other phase), the pressure-equilibrium projection, the finite-rate thermal and mass-transfer relaxation, and nucleation.

At a domain face f between the boundary cell N and its ghost G the fan decomposes the jump $q_G - q_N$ that the ghost fill created. For subsonic outflow exactly one wave enters cell N, the acoustic one; for subsonic reversed flow two enter, the acoustic wave and the contact, which carries the ghost's composition and entropy inward. By the time the level's fluxes are formed, the boundary cell's predicted increment is known to the last term — every interior face, the corrections, the transverse and viscous contributions, the non-conservative deposit — except for

the entering acoustic wave:

$$\Delta q_N = \underbrace{\Delta q^{\text{int}} + \Delta q^{f,\text{out}} + \Delta q^{f,\text{cont}}}_{\text{known}} - \frac{\Delta t}{\Delta x}\, s_{\text{ac}} \mathcal{W}_{\text{ac}}. \tag{8}$$

The construction below replaces $\mathcal{W}_{\text{ac}}$ and leaves everything else — including the ghost's contact wave in reversed flow — as the scheme produced it.

5 The construction

5.1 The modelled wave

The entering acoustic wave is taken to be the HLLC star jump on the N side of f for a prescribed pressure amplitude a [Pa]. Two speeds of the frozen fan enter here, and they are frozen quantities because the fan is: s_{ac} is the HLLC estimate of the speed of the acoustic wave entering the domain (on a high face $S_L = \min(u_N - c_{f,N},\, u_G - c_{f,G})$, the outer wave speed on the domain side), and $c_{\text{eff}} = |s_{\text{ac}} - u_{n,N}|$ is that wave's speed relative to the boundary cell, which equals $c_{f,N}$ whenever the domain side sets the estimate. The modelled wave is thus a frozen wave, as every wave the scheme propagates is; the relaxation acts on the cell afterwards, in the trial evaluation. With $n_{\text{sgn}} = \pm 1$ the outward sign,

$$S_M(a) = u_{n,N} - \frac{n_{\text{sgn}}\, a}{\rho_N\, c_{\text{eff}}}, \qquad \hat{\mathcal{W}}(a) = U_N^{\star}\big(s_{\text{ac}}, S_M(a)\big) - q_N, \tag{9}$$

so that the HLLC pressure behind the wave is exactly $P_N + a$ and the velocity jump is that of an inward-running acoustic wave, $\mathrm{d}u_{\text{out}} = -a/\rho_N c_{\text{eff}}$. The star state is built by the interior's own routine, so the wave satisfies the conservation identities the scheme relies on, refuses (rather than clamps) when the jump is inadmissible, and needs no new discretisation at the boundary. $a = a_{\text{fan}}$ reproduces the wave the ghost fan actually sent; $a = 0$ is no wave. The modelled wave enters first order — formally: on an outflow the entering wave is $O(r\,p')$ small, and the order series of §7 finds no trace of it.

5.2 The trial state

For a candidate a the boundary cell's end-of-step state and its relaxed image are

$$q^{\text{h}}(a) = q_N + \Delta q^{\text{known}} - \frac{\Delta t}{\Delta x}\, s_{\text{ac}} \hat{\mathcal{W}}(a), \qquad \tilde{q}(a) = \mathcal{R}\big(q^{\text{h}}(a);\, \Delta t_r\big), \tag{10}$$

where $\mathcal{R}$ is the production reaction chain applied to the boundary strip only, with its counters snapshotted so the trials leave no trace in the solution's accounting. From $\tilde{q}$ we read the six-equation mixture pressure $\tilde{P} = \alpha_1 P_1 + \alpha_2 P_2$, the outward velocity $\tilde{u_{\text{out}}}$ and the mixture density $\tilde{\rho}$. This is the Sutherland–Kennedy move — the stiff source inside the boundary relation — carried out by evaluating the source on trial states rather than by deriving a relaxed eigenstructure.

5.3 The condition

Let $I(q) = P(q) - Z\, u_{\text{out}}(q)$ be the incoming relaxed invariant of the boundary cell. In the LODI frame, $\partial_t(P - \rho c\, u_{\text{out}}) = -\mathcal{L}_{\text{in}}$ exactly, so every classical outflow is a statement about $\mathcal{L}_{\text{in}}$, and we discretise it on the relaxed step:

$$\text{rate form:} \quad I(\tilde{q}(a)) - I(q_N) + \Delta t\, K\,(P_N - P_\infty) = 0, \tag{11}$$

$$\text{value form:} \quad I(\tilde{q}(a)) - P_\infty = 0. \tag{12}$$

The rate form with $K = 0$ is Poinsot–Lele's $\mathcal{L}_{\text{in}} = 0$; with $K = \sigma(1 - M^2)c/L$ it is the Rudy–Strikwerda anchoring. The value form is Hedstrom's: the incoming invariant *equals* its far-field value every step. The two differ only when I has drifted from P_∞; the value form restores it within the step, the rate form never does. Both are non-reflecting for an outgoing wave that keeps I constant — which a relaxed wave does when Z is its impedance.

5.4 The impedance

Z was first taken as the Wood impedance $\rho_N c_w$ of the boundary cell. With the wave at the Wood speed (the pressure constraint only, the thermal and mass-transfer legs off) the condition is then exact to the single-phase level; with the production closure the wave travels at the equilibrium speed and a residual reflection of 0.07 remains, the Z_w–Z_{eq} gap. The gap is closed by measuring Z on the trials themselves: the two seed evaluations at $a = 0$ and $a = a_{\text{fan}}$ give relaxed states that differ by one injected frozen wave, and

$$c_{\text{eq}}^2 = \frac{\tilde{P}(a_1) - \tilde{P}(a_0)}{\tilde{\rho}(a_1) - \tilde{\rho}(a_0)}, \qquad Z = \rho_N c_{\text{eq}}, \tag{13}$$

is the equilibrium slope the reaction chain itself imposes on the cell, whatever the relaxation regime — no EOS derivative along the coexistence curve, no neighbour, no model of which legs are fast. A degenerate pair keeps the Wood value and is counted.

5.5 Face types

The face type is a deck choice per face, never a runtime predicate on the state. `nr` is the condition above. `vent` is an exit into a plenum at P_∞, and treats its two flow directions differently, as every characteristic condition distinguishes them by the entering waves. A cell is classed inflow or outflow by the sign of $\tilde{u_{\text{out}}}$ on the $a = 0$ trial, the cell's own relaxed evolution with nothing entering. An outflow cell takes as its entering wave the substepped fan of §3 — the boundary cell's phase states marched in pressure toward P_∞ along their isentropes with HEM re-closure inside the dome, terminated at P_∞ or at the critical-flow point — with the wave's amplitude fixed by the fan, no secant, the choke being the fan's own termination; the trial evaluation is still run on that wave, so the probe reports the relaxed face state after the fan's wave has entered. An inflow cell solves the Hedstrom relation (12), which is the linearised stagnation condition for fluid entering from a plenum at rest ($P < P_\infty$ by the dynamic head for inward u). Two earlier forms of the vent were built and refuted on the way to this one: a static-pressure Dirichlet $\tilde{P} = P_\infty$ solved by the secant with a choke re-solve $\tilde{u_{\text{out}}} = c_{\text{eq}}$ where the ambient solution was relaxed-supersonic, which over-drove the reversed-flow part of the outlet to supersonic inflow (§7.1) and could not represent the flashing expansion (§7); and the same with the plenum inflow relation, which fixed the inflow but left the outflow cells with a one-jump amplitude that had no root on the liquid branch on the flashing vent. The vent type exists because the production outlet needed it.

5.6 The scalar solve, refusal, cost

$g(a)$ is one scalar in one unknown, nearly linear (slope $\nu\,(c_{\text{eq}}^2/c_f^2 + c_{\text{eq}}/c_f)$ with ν the swept fraction), solved by a secant from the seeds $\{0, a_{\text{fan}}\}$ with a fixed number of updates and no tolerance; the residual is reported. Any evaluation that refuses — a star state with a negative partial mass, a trial state whose host phase has no EOS root (tested on the EOS's non-aborting surface with the same dispatch the flux uses), a non-finite result — drops the cell to the fan's own wave and is counted. Three to four reaction evaluations per boundary cell per step. There is no Mach dispatch anywhere: the frozen fan decides whether a wave enters, the condition decides its strength.

5.7 Transverse terms and corners

In 3D-NSCBC the boundary relation is $\mathcal{L}_{\text{in}} = K(P - P_\infty) + (1 - \beta)\mathcal{T}$. Here the transverse fluctuations of the boundary cell are part of the known increment (8), so the entering wave compensates them in full — $\beta = 0$ — without anything transverse being computed; $\beta > 0$ would need the transverse change of I separately. A corner cell is processed once per face in direction order.

6 Contrast with Poinsot–Lele

What is modelled. P&L model a wave amplitude in a differential relation and integrate a boundary ODE. We model the strength of a discrete wave the interior scheme already propagates; the outgoing waves are not modelled at all.

What the condition is posed on. P&L impose it on the frozen eigenstructure. We impose it on the state after the source has acted, with the source's own impedance. Sutherland–Kennedy with relaxation as the source, without the relaxed eigenstructure on paper.

When it is applied. P&L at the level of the PDE; we at the level of one time step with the complete increment of the boundary cell in hand. Transverse effects enter through that increment.

What the far field means. P&L's non-reflecting limit says the incoming invariant does not change; the far field enters only through K. Hedstrom, and the ghost construction we shipped, set the incoming invariant's *value* to its far-field value every step. On the production outlet the difference was decisive (§7.1).

What is deliberately absent. A Mach dispatch on any relaxed speed; any use of the ghost's invariants; any tolerance; any division by a wave speed — the near-sonic blow-up of ghost-cell CBC is absent by construction, its analogue being a conditioning loss of the secant as the entering wave's swept fraction vanishes.

7 What has been measured

r is reflected over incident pressure amplitude; 256 cells, a 1 m/s velocity pulse, both faces NSCBC.

case	ghost	relaxed, Wood Z	relaxed, trial Z	wave speed
N1 vapour 20 bar / 280 K	+0.0005	−0.00004	−0.00002	234 m/s
N1b mixture $\alpha_1 = 0.05$, full closure	+0.241	+0.067	**−0.011**	139 m/s
N1b, pressure constraint only	+0.144	+0.005	+0.005	174 m/s (Wood)
N2 2-D pulse, residual/A, 4/5/6 ms	.0083/.0067/.0071	.0091/.0072/.0056	same	—

The value form on N1 gives the ghost's +0.0005 (in vapour the two are the same condition) and on N1b +0.0052. N4, mean-pressure recovery from $1.1P_\infty$ through both faces, in units of the initial excess per millisecond: ghost $1, .763, .528, .293, .060, .008, \ldots$; relaxed `nr` (value) $1, .761, .526, .291, .058, .008, \ldots$; `vent` $1, .54, .08, -.37, -.82, -.73, -.51, -.29, -.07, .007$ (an open end that rings once and is then absorbed on the inflow phase). Every run was bit-identical to the shipped code with the construction off, and no trial ever had to fall back on the 1-D set.

The flashing vent (the 100 bar supercritical liquid tube of §3 venting to 1 bar, vented mass at 80 μs against the boundary-free reference): the shipped choked fan +4.4 %; relaxed `nr` −38 % (the frozen linear number); the secant `vent` +14 %, with the secant accepting four early steps at

residuals of half the pressure and refusing every step after; the `vent` with the fan as its outflow wave model reproduces the shipped +4.4 % to every digit with no refusal. The one-wave form does not do this case; the fan does, inside the relaxed construction.

Order at the face. N1 at 256/512/1024 cells against a boundary-free reference at the same Δx and Δt (the same grid extended so that no face reflection can return in the window), scoring the face cell's pressure error integrated over the pulse's passage: the shipped ghost falls by 4.2 and 3.4 per refinement, the relaxed boundary over the same ghost by 4.2 and 3.4 — equal to four digits — so the entering wave's dropped correction costs nothing measurable and the face is second order. The reflected residual left in the domain, 0.23 Pa on a 5 kPa pulse for every boundary at every resolution, does not converge: it is the amplitude nonlinearity of the linearised Hedstrom invariant (a 50 m/s pulse gives 40× the reflection of a 1 m/s one), not discretisation.

What the ghost still supplies. With the entering wave modelled and the vent's wave the fan's, the ghost state at an NSCBC face is read only for the outgoing waves at the face and, in reversed flow, the contact. Replacing the characteristic ghost by a plain copy of the boundary cell leaves N1, N4 and the 2-D pulse unchanged, moves N1b's reflection from +0.005 to −0.006 (the same magnitude, and within 10 % of the ghost on the boundary-free residual), costs 1.1 % of the vented mass at the flashing vent, and drops the face to first order on outgoing waves (ratios 1.8, 1.9; 5–22× the face-cell error): what a copy removes is the outgoing wave's second-order correction, not anything characteristic. Replacing it instead by a minmod continuation of the phase states past the face (the mixture assembled from the phases so that $\rho = m_1 + m_2$ and $\rho E = \mathcal{E}_1 + \mathcal{E}_2$ hold exactly; a continuation with no EOS root refused to the copy and counted) recovers the ghost's order and values to four digits on N1 and N4, −0.28 % on the flashing vent, and the smallest residual of the three on N1b. If that holds on the production outlet, the characteristic construction — invariants, extrapolation, Mach dispatch, strain cap — has nothing left to supply and is retired.

7.1 The production outlet

A saturated two-phase jet from a pipe rupture reaches the outflow plane 20 diameters downstream as an overexpanded supersonic core ($u/c \approx 1.5$, 13 bar in gas at 20 bar) with 60–70 % of the face in reversed flow. The shipped ghost relieves the arrival transient within a millisecond and lets the core leave supersonically. Each version of the relaxed construction was restarted from the same checkpoint before arrival and run through it.

Rate form, `nr`**.** Robust — no refusal on any face in 3000 steps, the validator clean — and wrong: the last cells hold 40–47 bar against a supersonic core behind them. The probe shows the ghost sending an expansion of −0.5 to −2 MPa per step at arrival and the relaxed condition choosing +40 kPa. $\mathcal{L}_{\text{in}} = 0$ carries no far field: once the arrival raised P and the flow reversed, $P - Zu_{\text{out}}$ stayed where it was.

Value form, `nr`**.** Worse, 46–58 bar and the shock ten cells inside by 24 ms, for a reason that took the numbers to see: at the arrival cell $P = 38$ bar, $u = 85$ m/s, $\rho c \approx 3.5 \times 10^4$, so $P - \rho c u \approx 8$ bar, *below* P_∞, and the Hedstrom invariant condition asks for a compression. The ghost's linear branch would too; the ghost relieves because the jump exceeds its strain cap and it switches to the choked fan, which ignores the invariants. The shipped face is a characteristic outflow at small jumps and a venting outlet at large ones, with a threshold deciding. An acoustic linearisation about a quiescent far field accepts $P \neq P_\infty$ compensated by velocity; an exit into a plenum does not.

`vent`**, Dirichlet on all subsonic cells.** The arrival is relieved (harder than the ghost: face at 6.5 bar, core through at $u/c = 1.95$) and the picture at +2 ms matches the ghost's. Later

it goes the other way: the reversed-flow part of the face is driven to supersonic *inflow* (u/c to -2.2), the mass pushed in recompresses the core and the shock walks inside. A static-pressure Dirichlet is the exit condition for outflow into a plenum; for inflow from a plenum at rest the condition is stagnation.

`vent` **as a plenum (inflow on the Hedstrom relation).** The arrival as before; the reversed-flow pathology gone (inflow u/c no lower than -0.16), the inflow cells sitting below P_∞ as the stagnation relation has them, the time step tracking the ghost's. The core takes a detour the ghost does not — subsonic at 22 bar at +4 ms where the ghost is supersonic at 14, supersonic again at +8 ms — a recompression excursion consistent with the open-end ring of the N4 test acting on the subsonic surroundings; whether that or the ghost's earlier settling is the better outlet is not decidable against the ghost, since both are boundary constructions. Two to three cells per step still refused to the fan's wave in the late phase.

`vent` **with the fan as the outflow wave model.** Identical to the previous run to the second decimal at every station; the refusals fall from 7979 in 3000 steps to 1, because the fan reaches the same relaxed state along an admissible path with a wave one tenth the secant's amplitude. This is the configuration that ships: `nr` on the lateral faces, `vent` on the outlet, the value form, the trial impedance.

Ghost fill: copy. With the characteristic ghost replaced by a copy of the boundary cell, the outlet is again identical to the previous run to the second decimal through the reversed-flow phase, with no refusal on any face: the plenum inflow relation on the cell, not the ghost's contact wave, is what sets the inflow. The one change is diagnostic — with no wave sent by the ghost the two seed trials of (13) coincide on a third of the lateral-face cells and the impedance falls back to Wood there, harmlessly (the cells are at rest near P_∞).

Ghost fill: continuation. A first form that continued every conserved slot separately was refuted by the validator on the first step (a hundred boundary cells with $\rho \neq m_1 + m_2$ at 10^{-9}) and, 2300 steps later, by a time-step collapse and an abort; the assembled form of §7 is running at the time of writing.

8 Questions for discussion

1. **Value or rate.** Hedstrom's value condition on the relaxed invariant against Poinsot–Lele's rate condition with a K. On plane waves both are transparent; on the impinging outlet the value form anchors strongly and still could not vent. Is there a principled way to choose the anchoring strength for a face that sees both small-amplitude acoustics and an order-one transient, other than a threshold on the jump?

2. **The source along the wave.** For a flashing expansion the relaxation acts along the fan, not after the wave has entered the cell, and a frozen star jump cannot represent it. Is there an analogue in reacting flow — a case where the reaction must be integrated along the characteristic rather than in the boundary relation — and how was it treated? Our answer is to keep the substepped fan as the outflow wave model of the vent type and use the trial evaluation only to read the relaxed result; it reproduces the flashing-vent benchmark exactly and the production outlet to the second decimal.

3. **Reversed flow.** On the part of the face where the flow enters, the acoustic wave is the modelled one and the contact wave carries whatever the ghost holds. What should enter: the cell's own fluid, or the far-field composition and temperature? On our outlet the answer made no measurable difference — a ghost that sends no contact at all reproduces

the outlet, because the plenum inflow relation on the cell sets the inflow — but the far field there has the composition of the jet. In reacting flow this is the burnt-gas backflow question; the answer probably carries over.

4. **The impedance from the trials.** Equation (13) measures the equilibrium slope the source imposes over one step and adapts to whatever the relaxation regime is. Is there a cleaner statement of what invariant a relaxing system conserves across a subsonic outflow, so that Z need not be measured?

5. **Transverse terms.** Compensating the boundary cell's transverse fluctuations in full ($\beta = 0$) through the known increment left the 2-D pulse's corner residual unchanged at about 6 %. Is the Yoo–Im $\beta = M$ compromise still the recommendation when the transverse terms are known exactly rather than estimated?

6. **Order and conditioning.** The modelled wave enters first order (no limited correction for a wave with no upwind neighbour), yet the measured order at the face is the ghost's, second, because the entering wave on an outflow is small; what does cost an order is a ghost that carries no outgoing wave (a copy). Masset et al.'s bound — the minimum over the extrapolation, the outgoing-wave estimate and the scheme — reads the same way. Is there a case in reacting flow where the entering wave is large enough for its order to matter, and what was done there?

7. **The relaxed-supersonic reading.** The frozen fan decides whether a wave enters; when the flow is relaxed-supersonic but frozen-subsonic the condition is still imposed. A dispatch on the relaxed Mach number was refuted on this outlet (it held the jet head against the plane). Is there a better statement for that regime?

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